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June 19, 1964
SUPERSEDING
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January 29, 1951

FEDERAL SPECIFICATION

BRUSH, FILE CLEANER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers requirements for one type file cleaner brush which is a combination card, brush, and scorer.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

H–S–951—Synthetic Filaments, General Specification for Brushes.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–W–6110—Wood, Determination of Moisture Content of.

MIL–H–15424—Hand Tools, Packaging of.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), before production is commenced, a sample of the item covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product. The brushes delivered under this specification shall be the manufacturer's standard commercial product, except for any modification and additional requirements specified herein.

3.3 Materials. The materials used shall conform to the requirements specified hereinafter. Materials not definitely specified, shall be of the quality normally used for the purpose in commercial practice.

3.3.1 *Wood.* The wood shall be close-grained and straight-grained beech, birch, or

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maple. The moisture content, at time of fabrication, shall be not less than 8 percent or more than 12 percent when tested as specified in 4.3.2.

3.3.2 *Card wire.* The card wire shall be steel with a nominal diameter of 0.010 inch.

3.3.3 *Filler material.* Filler material shall be either tampico fiber or brass wire or synthetic filament as specified (see 6.2).

3.3.3.1 *Tampico fiber.* The tampico fiber shall be medium grade, natural and unbleached (see table II).

3.3.3.2 *Brass wire.* The brass wire shall be half-hard and shall have a nominal diameter of 0.009 inch.

3.3.3.3 *Synthetic filament.* The synthetic filament shall be black or natural color nylon with a minimum diameter of 0.010 inch and shall meet the requirements for class N, type II filament of H–S–951.

3.4 Construction. The brush shall be constructed similar to figure 1 and shall conform to the requirements specified herein and in table I. The brush shall consist of a wooden block with a card side and a filler side. The brush shall have either a wooden or steel handle and a detachable scorer. The filler material and card wires shall not show permanent bend or leave noticeable deposits of metal on a file when tested as specified in 4.4.1.

Table I—*Minimum physical requirements*

Block:	
Length (including handle)—inches	9
Width—do	2
Thickness—inch	½
Filler side of blocks:	
Diameter of holes—do	¼
Number of rows, width direction	4
Number of rows, length direction	11
Length of filler clear of block—inch	⅝
Width of filler section—inches	1⅜
Length of filler section—do	4½
Weight of filler material:	
Tampico fiber—ounce	¾
Brass wire—ounces	2½
Synthetic filament—ounce	⅝
Card side of block:	
Width of wire section—inches	1½
Length of wire section—do	4½
Length of wire clear of backing—inch	¼
Number of wires per square inch	300
Handle: Length—inches	3½
Scorer: Length—do	3½

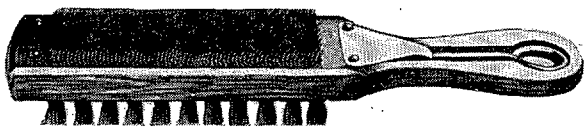


FIGURE 1. *File cleaner.*

3.4.1 *Block.* The block shall be of wood specified in 3.3.1. The corners shall be rounded and there shall be no sharp edges. The block shall be recessed to receive the backing of the card. The backing shall be flush with the top of the block. The block shall be recessed on the card side for the flattened end of the scorer.

3.4.1.1 *Card side.* The card side of the block shall be constructed of wires specified in 3.3.2, permanently fixed to either a canvas or leather backing at the option of the manufacturer. The wires shall pass through the backing and shall be inclined in the direction of the handle. The canvas backing shall be secured to the block with glue and No. 20 flathead, steel wire nails, nominal ⅜-inch long. The leather backing shall be secured to the block with not less than four steel, flathead, wood screws. The card wires shall not fall out or show a permanent bend when tested as specified in 4.4.1.

3.4.1.2 *Filler side.* The filler side of the block shall consist of filler material specified in 3.3.3, either staple-set or pitch-set at the option of the manufacturer, into holes in the wooden block. The filler material shall not come out of the block when tested as specified in 4.4.2. The weight of the filler material shall be as required in table I, when tested as specified in 4.4.3.

3.4.2 *Handle.* The handle shall be either wood, integral with the block, or shall be steel at the option of the manufacturer.

3.4.2.1 *Wood.* Wood handles shall be recessed on the card side of the block for accommodating the scorer, and shall have a ¼-inch-minimum diameter hole concentric with the circular loop of the scorer for hanging the brush.

3.4.2.2 *Steel.* Steel handles shall be of the open center type, fabricated of cadmium- or nickel-plated steel. The steel shall be a minimum of ⅜-inch wide and of a thickness which will prevent bending in use. The steel shall inclose the lengthwise edges of the wooden block and have a handgrip handle portion.

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3.4.3 *Scorer.* The scorer shall have a minimum diameter of 0.010 inch and shall be either soft steel or iron wire at the option of the manufacturer. The scorer shall be flattened on one end for a minimum of ⅜-inch and the opposite end shall have a circular loop with a minimum outside diameter of ½-inch. The scorer shall be secured in the recess by a metal spring securely fastened to the block. There shall be no movement of the scorer once in place. On steel handles, the circular loop of the scorer shall be located between the sides of the handle.

3.5 **Finish.** The wood shall have all rough edges removed and shall be finished in accordance with the manufacturer's standard commercial practice.

3.6 **Marking for identification.** In addition to any special marking required by the contract or order (see 6.2), the supplier shall permanently indent his name or trademark of such known character easily identifiable with the manufacturer, on the wooden block of each brush.

3.7 **Workmanship.** The finished brush shall conform to the grade and quality established by this specification. The occurrence of defects shall not exceed the applicable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Preproduction sample inspection.** The preproduction sample shall be examined in accordance with defects specified in 4.3.3.1 and specified dimensions, and shall be tested in accordance with 4.4.1 through 4.4.3.

4.3 **Sampling and inspection for acceptance.** Sampling and inspection shall be performed in accordance with the provisions set forth in MIL–STD–105, except where otherwise indicated.

4.3.1 *Inspection of components and materials.* In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested and inspected in accordance with the specified requirements of referenced subsidiary specifications and standards or, if none, in accordance with this specification.

4.3.2 *Intermediate testing.* Test shall be performed on each sample block at time of fabrication for moisture content in accordance with MIL–W–6110 to determine compliance with 3.3.1. Two determinations shall be taken on each sample unit and results shall be the average of the two determinations. The inspection level shall be S–2 with an acceptable quality level (AQL) of 2.5 defects per hundred units.

4.3.2.1 *In-process examination.* In-process examination shall be made to determine compliance with features which cannot be examined at time of end item inspection. These examinations shall apply to such characteristics as fixing of file card wires into leather or canvas backing (see 3.4.1.1); securing of canvas or leather backing to the block (see 3.4.1.1), uniformity and diameter of holes in wooden block (see 3.4.1.2); securing of filler material (staple-set or pitch-set) into holes in the block; any split, check, rot or other defect which may affect serviceability in filler side of wooden block. Any deviation from specification requirements shall be corrected. Failure to make immediate correction shall be cause for rejection of affected end item lot.

4.3.3 *End item inspection.* All the brushes offered for acceptance at one time shall be considered a lot for the purpose of inspection. The sample unit for this inspection shall be one completely fabricated brush.

4.3.3.1 *Visual examination.* Examination of the end item shall be in accordance with the list of defects set forth. The inspection level shall be level I and the acceptable quality level (AQL) shall be 4.0, expressed in terms of defects per hundred units.

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Examine	Defect
Construction and workmanship.	Part missing or not specified type. Fractured, split, dented, or malformed part. Part misplaced, loose, or not in proper alignment. Sharp burr, edge, sliver, or splinter.
Wooden block	Corners not rounded or sharp edges. Card backing not flush with face of block. Not recessed for scorer.
Card side	Not inclined in direction of handle. Number of wires less than specified. Canvas backing not secured as specified (when applicable). Leather backing not secured with 4 screws (when applicable).
Filler side	Not proper number of rows.
Handle, wood (when applicable).	Not integral with block. Not recessed for scorer as specified, hole missing or not concentric with scorer loop.
Handle, steel (when applicable).	Lengthwise edges of block not fully inclosed. No handgrip portion.
Scorer	Not flattened at one end with circular loop at opposite end. Not detachable. Not attached as specified.
Finish	Missing, not as specified.
Marking	Identification marking omitted, incomplete.

4.3.3.2 *Dimensional examination.* Inspection shall be made of the end item for compliance with dimensions specified. Any departure from specified requirements shall constitute a defect. The inspection level shall be S–3 and the AQL shall be 6.5 expressed in terms of defects per hundred units.

4.3.3.3 *Testing of the end item.* The end item shall be tested for the applicable characteristics as indicated in table II herein, from each lot presented for acceptance. The sample unit shall be one brush and the lot size shall be expressed in units of brushes.

4.3.4 *Examination of preparation for delivery.* Inspection to determine compliance with preparation for delivery requirements shall be in accordance with MIL–H–15424.

4.4 Tests

4.4.1 *Cleaning test.* A medium cut mill file shall be used for filing soft steel or brass for a period of not less than 5 minutes. At the completion of the filing, the filler side and card side shall be used to clean the file. The file shall be examined to determine compliance with 3.4 and 3.4.1.1.

4.4.2 *Filler material retention.* Individual strands of filler material shall be held between the thumb and forefinger and pulled to determine compliance with 3.4.1.2.

4.4.3 *Weight of filler material.* The filler material shall be removed from the brush, conditioned for not less than 16 hours in air, at a relative humidity of 50 percent plus or minus 4 percent and at a temperature of 23.0° C. plus or minus 1.1° C. and weighed, in the same atmosphere, to determine compliance with table I and 3.4.1.2.

5. PREPARATION FOR DELIVERY

Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.

5.1 Packaging and packing. Packaging and packing for levels A, B, or C shall be in accordance with MIL–H–15424, except that only fiberboard containers shall be used for exterior shipping containers (see 6.2).

5.2 Marking

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military requirements.* Marking shall be in accordance with MIL–H–15424.

6. NOTES

6.1 Intended use. The brush covered by this specification is intended primarily to be

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Table II—*Instructions for testing*

Characteristic	Specification reference		Requirements applicable to—		Number determinations per unit	Results reported as—		Inspect level	AQL
	Requirement	Test method	Individ unit	Lot aver		Pass or fail	Numerically to nearest—		
Cleaning test	3.4 and 3.4.1.1	4. 4. 1	X		1	X		S-2	2. 5
Filler retention	3.4.1.2	4. 4. 2	X		10	X		S-2	2. 5
Weight of filler material	Table I and 3.4.1.2	4. 4. 3		X	1		0.1 ounce	S-2	2. 5
Identification of Tampico fiber.	3.3.3		Certificate of compliance						

used to remove filings and other matter from the teeth of files.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Whether preproduction sample is required (see 3.1).

(*c*) Filler material required (see 3.3.3).

(*d*) Additional identification marking, if required (see 3.6).

(*e*) Selection of applicable levels of packaging and packing (see 5.1).

6.3 Illustration. The illustration identified herein as figure 1 is for the purpose of identification and is not intended to preclude any brush which otherwise meets the requirements of this specification.

6.4 Transportation description. Transportation description applicable to this item is:

Rail:

Brushes, not otherwise indexed by name, carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Brushes not otherwise indexed by name. Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

CUSTODIANS:
Army—GL
Navy—S&A
Air Force—MOA

Interested activities:
Review
Army—GL, MO
Navy—S&A, Y, Sh
Air Force—MOA

Preparing activity:
Army—GL

User
Army—EL
Navy—MC

Copies of this specification may be purchased for 5 cents each.

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